

Reproduction

Asexual reproduction: A process resulting in the production of genetically identical offspring from one parent

Advantages of asexual:

- * No pollinators
- * One parent
- * Fast
- * Adapt to parent environment
- * Good characteristics are conserved

Disadvantages:

- * Offspring is more susceptible to diseases
- * Offspring cannot adapt to changes in environment
- * Competition
- * No evolution
- * No variation.

Examples of asexual reproduction:

- 1- Fungi by spores
- 2- Bacteria by mitosis
- 3- Potato by tubers
- 4- Onion by bulb
- 5- Strawberry by runners

Sexual reproduction: A process involving the fusion of the nuclei of two gametes to form a zygote. Offspring is genetically different from each other & parents

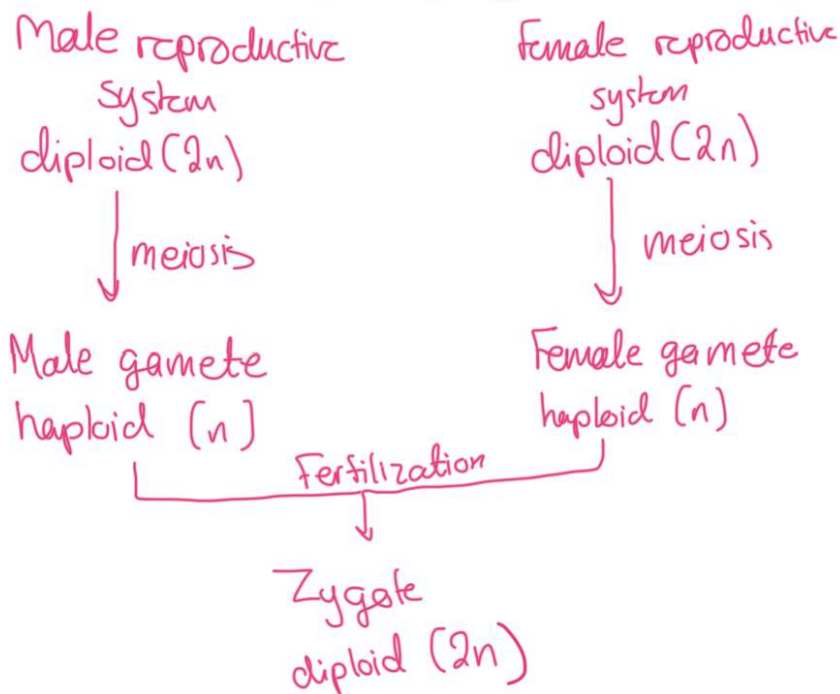
Advantages of sexual reproduction:

- * More variation
- * More evolution
- * Can adapt to changes in the environment
- * less competition between offspring
- * less susceptible to disease.

Disadvantages:

- * Requires two parents
- * Requires pollinators
- * Slower
- * More energy is required
- * less adapted because of variation

→ Number of chromosomes in all body cells is diploid ($2n$), has 2 sets of chromosomes, except gamete cells, which are haploid (n)



Two types of cell division:

1-Mitosis: Nuclear division giving rise to genetically identical cells

Roles of mitosis:

- * Repair, to replace damaged cells
- * Growth, increase in size by increasing number of cells
- * Asexual reproduction.

→ Chromosomes replicate before mitosis, then separate, maintaining the chromosome number in each daughter cell

Properties of mitosis:

- * Produces genetically identical cells.
- * Produces two daughter cells
- * Daughter cells have the same number of chromosomes
- * All body cells reproduce by mitosis, except the production of gametes

2- Meiosis: Reduction division in which the chromosome number is halved from diploid to haploid, resulting in genetically different cells.

Role of meiosis: Production of genetically different gametes, so increase genetic variation.

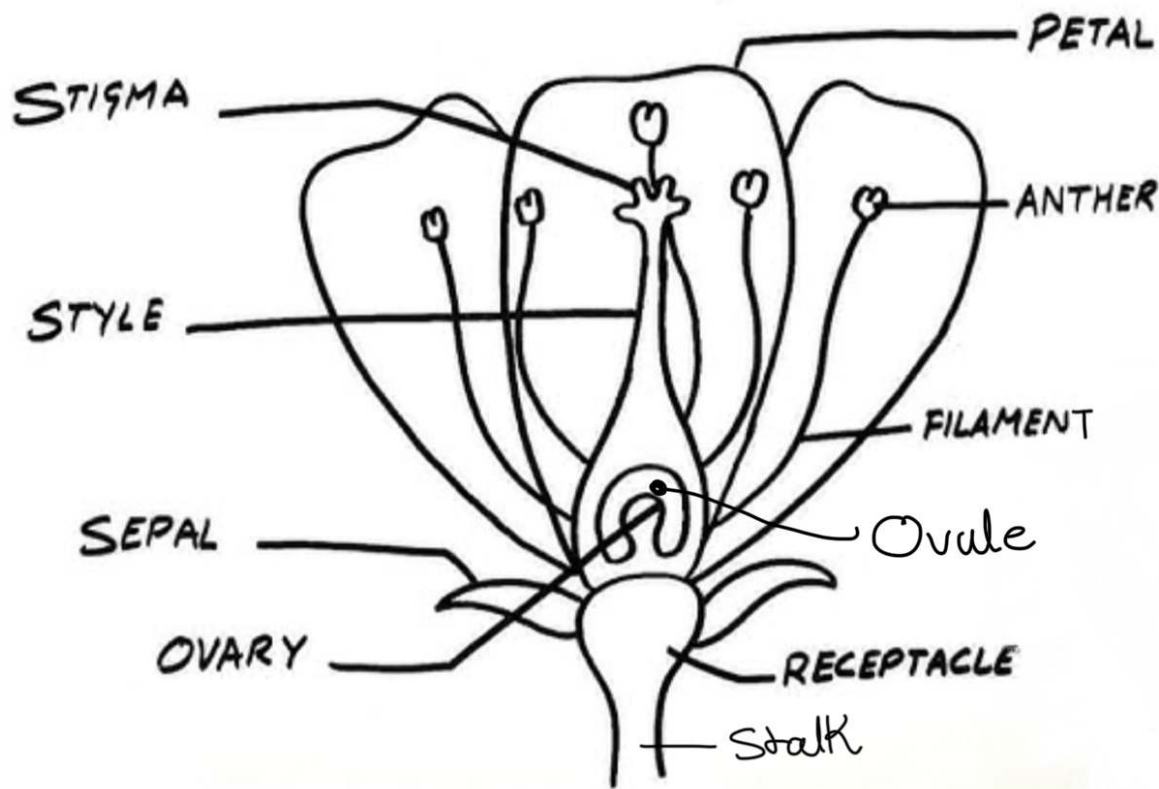
→ Chromosomes replicate before meiosis occur

Properties of meiosis:

- 1- Genetically different cells are produced
- 2- Daughter cells have half the number of chromosomes, so from diploid to haploid
- 3- Produces four daughter cells
- 4- Only used in the production of gametes.

* Haploid nucleus: A nucleus containing one set of chromosomes

* Diploid nucleus: A nucleus containing two sets of chromosomes



Functions:

- 1- Petal: Large and colorful to attract insects
- 2- Sepal: Protects flower bud
- 3- Stamen: (consists of anther & filament): Male reproductive part
- 4- Anther: Contains pollen sacs in which pollen grains are formed. Pollen contain males sex cells
- 5- Filament: Supports the anther
- 6- Carpel: Consists of stigma, style and ovary: female reproductive part
- 7- Stigma: Sticky surface that receives pollen during pollination
- 8- Style: Links the stigma to the ovary through which the pollen tube grows
- 9- Ovary: Contains ovule
- 10- Ovule: Contains haploid nucleus, which develops into a seed when fertilized

Steps of plant reproduction:

- 1- Pollination
- 2- Fertilization
- 3- Dispersal
- 4- Germination

Pollination: Transfer of pollen grains from anther to stigma

Types of pollination:

* Self-pollination: The transfer of pollen grains from the anther of a flower to the stigma of the same flower, or a different flower on the same plant.

Advantages of self-pollination:

- 1- Prevents extinction
- 2- More chances of fertilization
- 3- More chances of pollination
- 4- Parent plants are adapted to environment
- 5- Useful if plants are isolated

Disadvantages:

- 1- Cannot adapt to change in environments
- 2- Less genetic variation
- 3- More susceptible to disease

* Cross-pollination: The transfer of pollen grains from the anther of a flower to the stigma of a flower on a different plant of same specie.

Advantages of cross-pollination:

- 1- Allows variation
- 2- Plant is more likely to survive changes in environment
- 3- Plant will be resistant to disease
- 4- Chances of evolution.

Disadvantages:

- 1- Reliant on pollinators
- 2- Wastage of pollen
- 3- More energy is required
- 4- Need to produce more pollens
- 5- Risk of less fertilization

Methods of pollination:

Insect-pollinated

Adaptations of pollen:

- 1- Stickier
- 2- Heavier
- 3- Spikier
- 4- Larger

Adaptations of plant:

- 1- Petals are large, colored and scented
- 2- Have nectar to attract insects
- 3- Stamen is inside the flower
- 4- Stigma has small surface area, is inside the flower

Wind-pollinated

Adaptations of pollen:

- 1- Smaller
- 2- Lighter
- 3- Smoother
- 4- Not sticky

Adaptations of plant:

- 1- Petals are absent, if present they're small & green
- 2- Nectar is absent
- 3- Long filaments allowing the anther to dangle outside the flower
- 4- Stigma is large and feathery, dangles outside the plant.

Fertilization: When the pollen nucleus fuses with the ovule nucleus, forming diploid zygote.

Steps:

- 1- Enzymes from pollen grains digest the tissue down the style
- 2- Pollen tube grows down the style to the ovary, and attaches to the ovule at micropyle
- 3- Male gamete nucleus travels down the pollen tube, and enters the ovule
- 4- The male gamete nucleus fuses with the female gamete nucleus, forming a diploid nucleus.
- 5- Zygote divides by mitosis, forming an embryo, then the seed is formed
- 6- Ovary becomes bigger and ripe, forming the fruit
- 7- All other parts of the plant fall down.

Germination: Seeds develop into seedlings

Conditions for germination:

1- Optimum temperature for enzymes

2- Oxygen, to respire and release energy

3- Water, the swelling of seeds to activate enzymes.

→ Stored food in the seed is starched

→ Enzyme used is amylase

→ Mass of the seed gradually decreases, because stored food gets used to produce energy by respiration, as starch is broken down to glucose

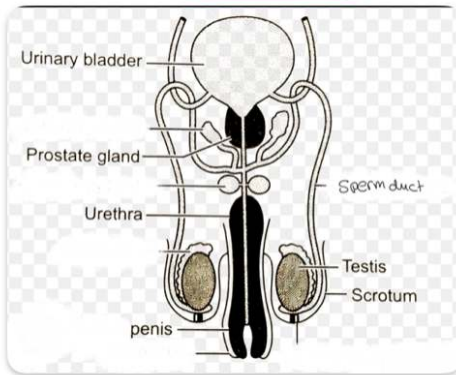
→ Mass of the seed may increase if water is absorbed, causing the seed to swell up

Human Reproduction: Sexual Reproduction

Male reproductive System

Organ: Testis
Diploid ($2n$)
↓ meiosis

Gamete: Sperm
haploid (n)



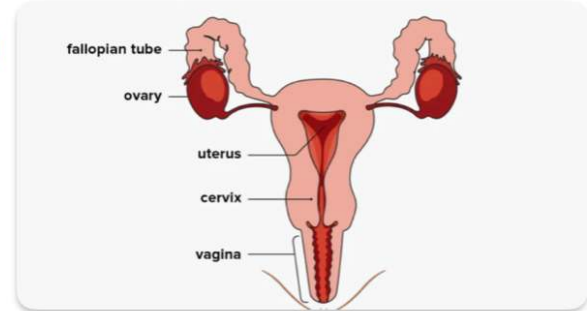
Functions:

- Testis: Male gonad that produces sperm
- Scrotum: A sac that holds the testis outside the body, keeping them cooler than body temperature.
- Sperm duct: Muscular tube that links the testis to the urethra to allow the passage of semen containing sperm
- Prostate gland: Adds fluids & nutrients to sperm to form semen
- Urethra: Passes semen through penis
- Penis: Becomes firm to insert into the vagina of the female during sexual intercourse in order to transfer sperm

Female reproductive system

Organ: Ovary
Diploid ($2n$)
↓ meiosis

Gamete: Ovum
haploid (n)



Functions:

- Ovary: Contains follicle in which ova is produced. (Female gonad)
- Oviduct: Carries the ovum to the uterus, with the propulsion provided by tiny cilia in the wall. Site of fertilization
- Uterus: Where the fetus develops
- Cervix: a ring of muscles that separates the vagina from the uterus
- Vagina: Receives male penis during sexual intercourse, where sperms are deposited

Fertilization

Zygote (diploid ($2n$))

↓ mitosis

Embryo ($2n$)

↓ Growth

Fetus ($2n$)

At the age of puberty, hormones are secreted, and when they circulate around the body, they cause the development of secondary sexual characteristics

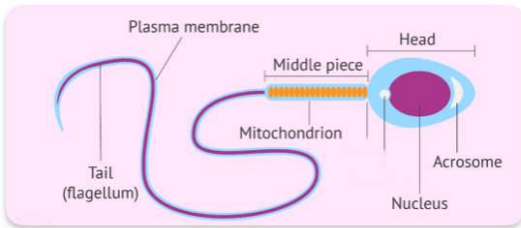
Males: By testosterone

- * Enlargement of testis and penis
- * Deepening of voice
- * Growth of hair in pubic region, face & armpits

Females: By oestrogen

- * Increased growth of breasts
- * Widening of hips
- * Growth of hair in pubic area & armpits
- * Increase in size of the uterus and vagina

Male gamete: Sperm



Functions:

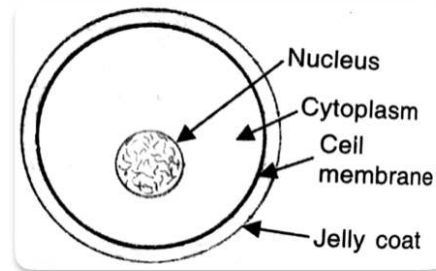
- Acrosome: Contains enzymes that digest the jelly coat of egg cell
- Nucleus: Contains chromosomes
- Flagellum: Allows the sperm to swim to the egg
- Mid-section: Has mitochondria to do respiration to release energy.

Properties:

- Produced in large numbers because many do not survive
- Produced at puberty
- Production of sperm is very sensitive to heat, that is why the testis is outside the body, because it is cooler
- Much smaller than egg

↑ motile
M. M. M
↓ mini ↓ many

Female gamete: Ovum



Functions:

- Jelly coat: Protects the contents of the cell and prevents more than one sperm from entering and fertilizing the egg. Hardens after fertilization
- Nucleus: Contain chromosomes
- Cytoplasm: Rich in fats & proteins:
Fats: Act as energy stores
Protein: For growth of egg if fertilized

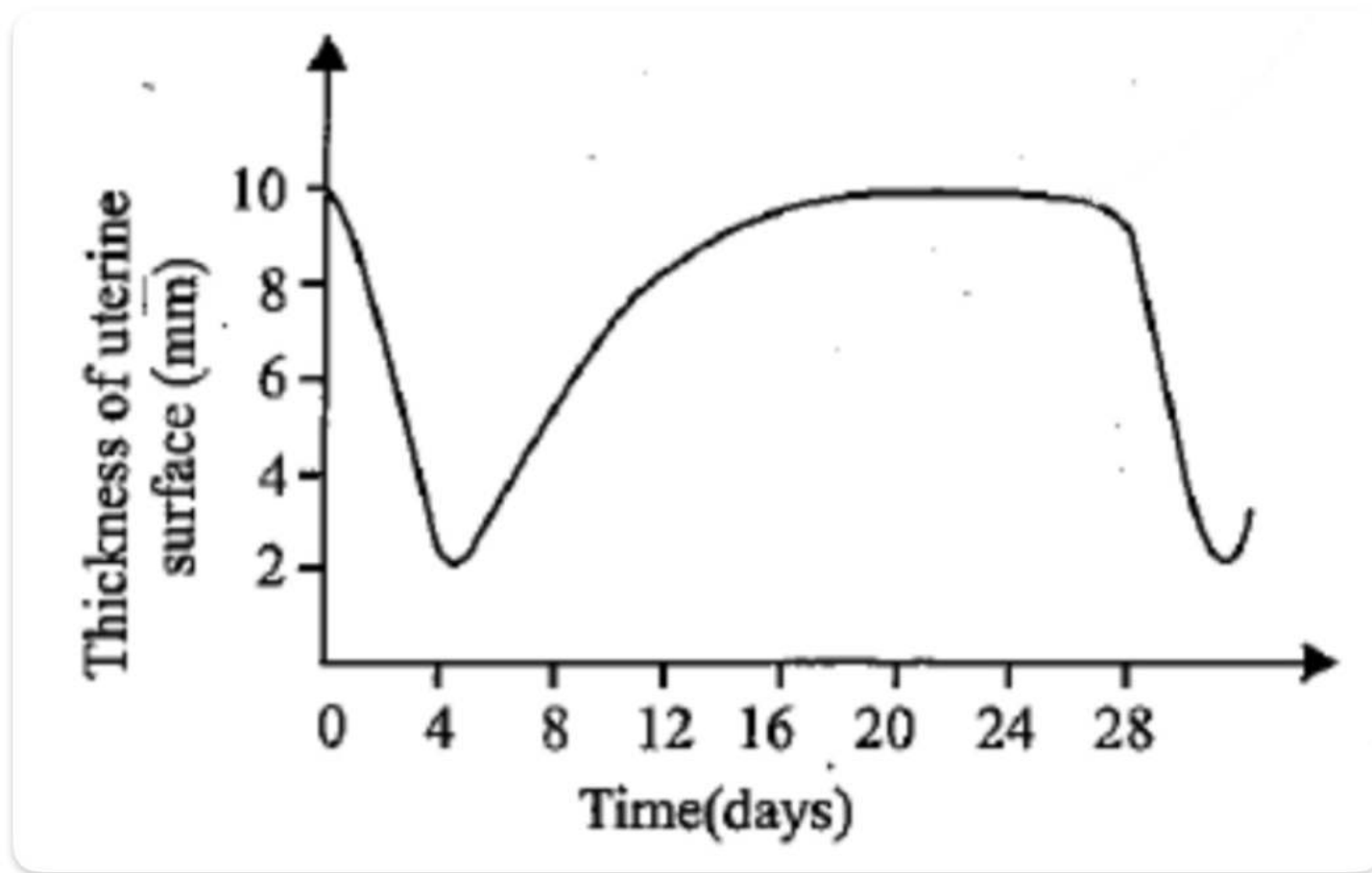
Properties:

- Much bigger than sperm
- Only one is produced each month
- Mature at puberty

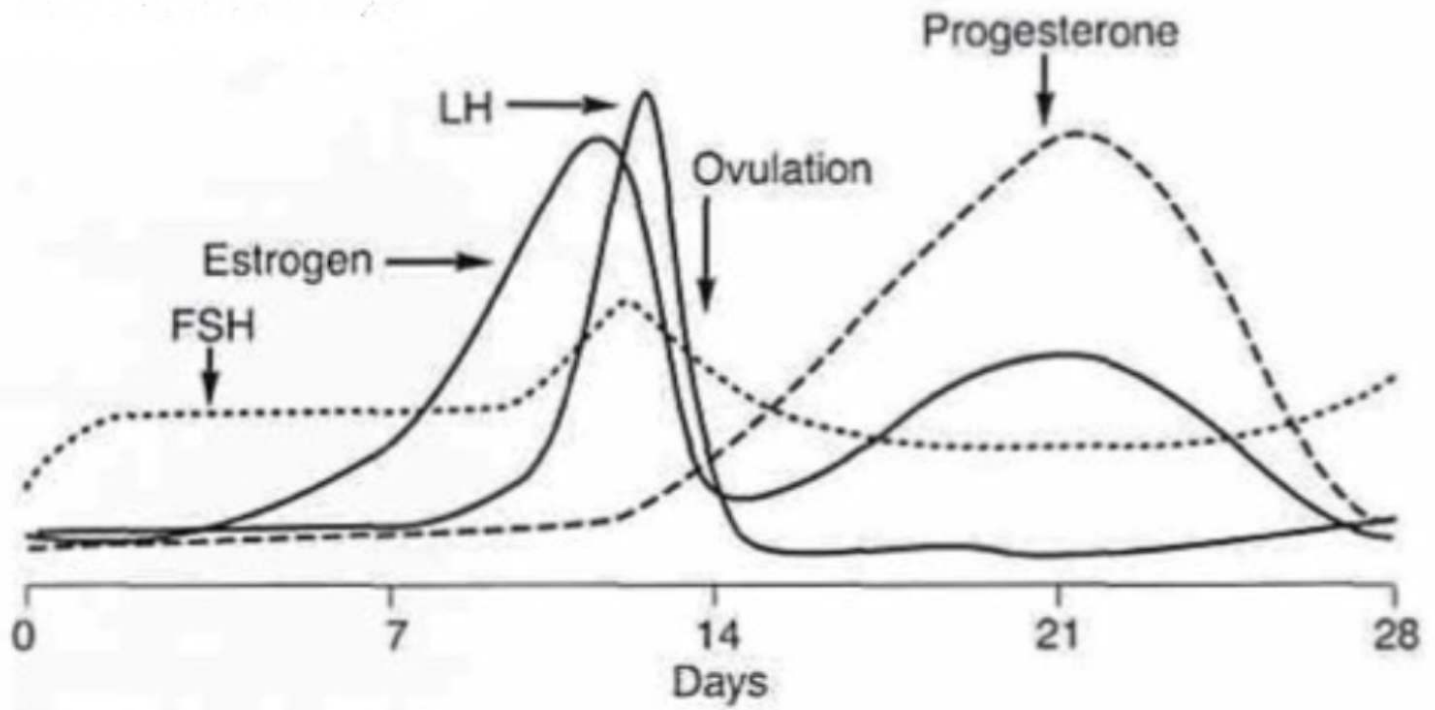
↑ Fixed
F. F. F
↓ fat ↓ few

Menstrual Cycle

- ⊗ FSH: A hormone secreted from the pituitary gland in the brain. FSH travels through the blood to the ovary. Stimulates the development of follicle that contains egg
- ⊗ Developing follicle in ovaries secretes oestrogen: A hormone that increases the thickening of the lining of the uterus.
- ⊗ Oestrogen travels by the blood to the pituitary gland, and stimulates it to release LH
- ⊗ LH: A hormone secreted by the pituitary gland when the follicle is fully developed. LH stimulates the follicle to burst open, releasing the ovum into the oviduct
- ⊗ Remaining of the follicle, called corpus luteum, secretes the hormone progesterone. Progesterone has the same function as oestrogen; increasing the thickness of the lining of the uterus
- ⊗ Progesterone also blocks FSH & LH, as well as maintaining the thickness of the lining of the uterus.
- ⊗ If the egg is not fertilized, the corpus luteum breaks down and the secretion of ^{progesterone} ~~oestrogen~~ stops. Lining of the uterus starts Shedding, and leaves through the vagina. Menstruation occurs



First 7 days of the cycle, the thickness of the lining of the uterus decreases. From 7 to 14 days, the thickening of the lining of the uterus increases. After ovulation, last two weeks, thickness of the lining of the uterus remains constant.



FSH: Secreted from pituitary gland

Oestrogen: Secreted from ovaries

LH: Secreted from pituitary gland

Progesterone: Secreted from ovaries

Ovulation: Release of ovum from follicle into the
Oviduct / Fallopian tube by the hormone LH.

Menstrual cycle: 28 days

Menstruation: 3-7 days

Ovulation: Middle of menstrual cycle

Pregnancy

① Fertilization:

- After ovulation, the egg is swept along the oviduct through cilia and peristalsis.

- Intercourse should happen from 2-3 days after ovulation in order for fertilization to happen, as the egg can live only 24 hours after ovulation.

The process of fertilization:

1- Enzyme is released from the acrosome in the sperm will digest the jelly coat surrounding the ovum

2- The head of the sperm enters the egg. The haploid male nucleus fuses with the haploid female nucleus to form a diploid zygote in the oviduct.

3- The diploid zygote divides several times by mitosis, forming a ball of cells called the embryo.

② Implantation:

1- The embryo moves through the oviduct to the uterus by ciliated cells and peristalsis.

2- The embryo implants itself in the lining of the uterus.

3- The placenta is formed, and connected to the embryo through the umbilical cord.

Egg cell $\xrightarrow[\text{Fertilization}]{+\text{Sperm}}$ Zygote $\xrightarrow{\text{Mitosis}}$ Embryo $\xrightarrow{\text{Growth}}$ Fetus

1- Placenta: Functions:

- * Prevents the mixing of maternal blood and fetal blood
- * Transport oxygen and nutrients from mother to fetus
- * Transport of carbon dioxide and urea from fetus to mother
- * Secretes progesterone hormone, which maintains the thickness of the lining of the uterus
- * Maintains a constant temperature of the fetus
- * Transport of antibodies from the mother to the fetus, giving the fetus with passive immunity

→ Some pathogens and toxins: Drugs, alcohol, nicotine, can pass through the placenta and affect the fetus

2- Umbilical cord: Function:

- * Connects the placenta to the fetus

3- Amniotic Sac: Functions:

- * Protects the fetus, and prevents the entrance of any pathogen
- * Secreting amniotic fluid.

4- Amniotic fluid: Functions:

- * Prevents the dryness of the fetus
- * Allows the movement of the fetus easily
- * Maintaining constant temperature of the fetus
- * Protects the fetus from any mechanical shocks

Sexual transmitted Infections (STI)

AIDS: Acquired immune Deficiency Syndrome

Caused by a virus called HIV: Human immunodeficiency virus
Has no treatment as it is caused by a virus.

Ways of transmission:

- Sexual intercourse: As a result of mixing bodily fluids
- Blood transfusion
- Sharing contaminated needles
- Through placenta and breastfeeding

HIV: * Damages the lymphocytes



* Less lymphocytes in the body



* Weak immune System



* Body is vulnerable to any disease infection

Symptom of AIDS in early stage is flu